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REVIEW ARTICLE

Paracentral acute middle maculopathy—review of the literature

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Abstract
Paracentral acute middle maculopathy (PAMM) is a recently identified spectral-domain optical coherence tomography (SD-OCT) finding characterized by a hyper-reflective band spanning the inner nuclear layer (INL), which typically evolves to INL atrophy in later stages. Typical clinical features include the sudden onset of one or multiple paracentral scotomas, normal or mild reduction in visual acuity, and a normal fundus appearance or a fundus with a deep grayish lesion. Although its pathophysiology is not yet fully understood, ischemia at the level of the intermediate and deep capillary plexa has been demonstrated to play a major role. Since its first description, an increasing number of publications on PAMM have been published in ophthalmology scientific journals. The purpose of this study is to provide a review of the current literature on PAMM.

Keywords Paracentral acute middle maculopathy · Deep capillary ischemia · Spectral-domain optical coherence tomography, optical coherence tomography angiography · Retinal imaging

Introduction

Paracentral acute middle maculopathy (PAMM), originally described in 2013 by Sarraf et al. [1], is characterized by a hyper-reflective band-like lesion involving the inner nuclear layer (INL), resulting in permanent INL thinning [1–5]. PAMM was initially considered a variant of acute macular neuroretinopathy (AMN), in which the lesions occur above the outer plexiform layer (type 1 AMN) [1]; however, PAMM is currently regarded as a different entity from AMN [2]. Of historical and pathophysiologic interest, middle retinal plaques were previously described in patients with retinal vein occlusions using optical coherence tomography (OCT) imaging in 2011 under the eponym of Paques' plaques [6]. However, the interest in macular hypoperfusion has grown with the development of spectral-domain OCT (SD-OCT) and OCT-angiography (OCT-A), which have allowed the localization of retinal vessel changes. Since its initial description, there has been an increasing number of reports and case series of patients with PAMM.

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Pathophysiology

The mechanism associated with PAMM is still a subject of active investigation [7–10], but there is a consensus that this is primarily due to sublethal ischemic hypoxia of the middle retinal tissue [1, 2, 4, 11]. OCT-A studies have reported changes in the intermediate (ICP) and deep capillary plexa (DCP) vasculature [12, 13]. The outer plexiform layer (OPL) and the INL are located in the so-called watershed zone [14] where oxygen is provided by both choroidal and retinal circulations [11, 14, 15]. When the oxygen supply starts to decline, the oxygen tension is highest near the superficial retinal layers and the choroid, rendering this area more sensitive to ischemia [11, 14]. The middle retina may be particularly susceptible to ischemic damage owing to the high-oxygen consumption by the horizontal cells [16]. The pathogenesis of PAMM may start at the ICP with secondary downstream changes in the DCP [7]. However, it has been contended that PAMM changes may rather reflect a state of middle retinal hypo-oxygenation arising from the inner retina, developing first at the level of the distal capillary circulation (ICP and DCP) [7, 8, 17, 18]. Although the SCP provides blood supply to the DCP via vertical anastomoses between the two plexa [19, 20], SCP involvement in PAMM is only occasional and only occurs concurrently with ICP/DCP hypoperfusion [7, 8, 18, 21, 22].

In cases in which the ischemia is not so severe, such as those in which the vascular flow is preserved or perfusion is rapidly restored, reperfusion injury may play an important role in the damage of the INL [5, 22, 23]. Reperfusion leads to increased formation of reactive oxygen species, nitric oxide neurotoxicity, inflammation, and extravascular leakage [21, 23]. Transient hypoperfusion of the DCP followed by immediate restoration of microcirculation may cause reperfusion injury and chronic changes of PAMM [5, 22]. Interestingly, however, one study found that cases in which reperfusion did not occur at the ICP showed more severe INL thinning in the chronic phase [7].

In patients with foveal hypoplasia (FH), a unique variant of PAMM has been reported, which was coined central acute middle maculopathy (CAMM) [24]. CAMM is characterized by acute changes in the INL like PAMM, progressing to a chronic phase in which inner and outer retinal layer degenerative changes occur [24]. In eyes with FH, there is disorganization of the vascular patterns of the deep and superficial retinal layers, which may be related to disease severity [19]. The foveal avascular zone (FAZ) is absent, and there is fusion of the SCP and DCP in the foveal center [18, 25, 26]. These changes lead to redistribution of the oxygen supply to the photoreceptors, with an increase in the relative contribution of the retinal circulation [24]. These changes may place the outer retinal layers at risk in the case of ischemia of this "single" foveal vascular layer; postischemic depletion of central Müller cells may also contribute to photoreceptor loss in this setting [24].

Independent from the etiology, PAMM lesions progress to atrophy of the INL [1, 2], which accounts for the persistent scotomata patients may report [3]. Figure 1 depicts the probable pathogenic mechanisms involving macular hypoperfusion and PAMM.

PAMM in retinal vascular diseases

PAMM has been associated with a number of retinal vascular diseases [2], providing further support that vascular dysfunction plays a pivotal role in PAMM. A number of publications have shown an association between retinal artery or venous occlusions (RVOs) and PAMM [10, 11, 27–31], including cilioretinal arterial occlusion [9, 32]. In the first and largest study of PAMM in eyes with nonischemic central retinal vein occlusion (CRVO) [14], twenty-five of the 484 patients diagnosed with CRVO had evidence of PAMM on SD-OCT [14]. Interestingly, one study found resolved PAMM lesions in the fellow eyes of patients with unilateral RVO, reinforcing the association between RVO and PAMM [33]. Purtscher's retinopathy has also been reported to accompany PAMM due to embolic or thrombotic occlusion [34–36]. PAMM has also been identified with other retinal vascular diseases, such as sickle cell retinopathy (SCR) [37] and inflammatory occlusive retinal vasculitis [38]. PAMM may be the precursor lesion leading to macular thinning in SCR as a result of acute ischemic events occurring at the DCP [15]. PAMM has also been described after cosmetic procedures, such as filler injections, as small filler materials enter the ophthalmic artery and occlude downstream arterioles and precapillaries [30, 36, 39]. Finally, both hypertensive retinopathy and diabetic retinopathy have been reported to be associated with PAMM [1, 4, 38, 40]. Even patients with mild systemic arterial hypertension (SAH) and low cardiovascular risk have been found to have small chronic PAMM lesions in up to 88% of cases; these changes may represent the earliest changes in retinal microcirculation that precede detectable quantitative changes in retinal vessel density [41].

PAMM in ocular diseases

A number of ocular conditions have been reported in association with PAMM, including inflammatory chorioretinopathies, congenital glaucoma [42], and FH [24]. PAMM in the setting of birdshot chorioretinopathy may be due to transient occlusion of the retinal capillary plexa [43]. PAMM has been reported in patients with juvenile dermatomyositis, likely due to vasculitic, occlusive retinopathy [44, 45].

PAMM has also been associated with both intraocular (phacoemulsification [46, 47], pars plana vitrectomy for

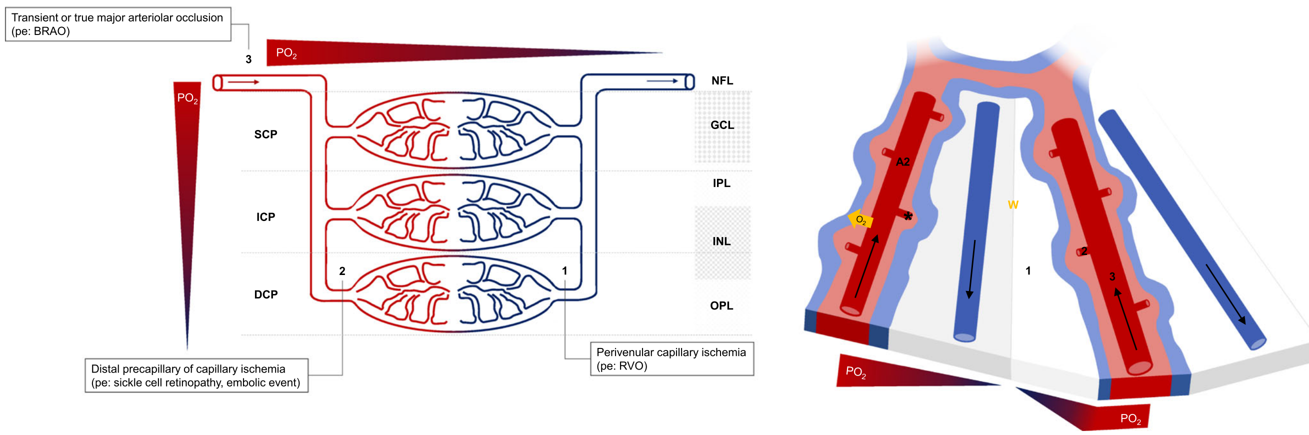

Fig. 1 Schematic representation of the middle retinal (MR) perfusion and pathogenesis of paracentral acute middle maculopathy (PAMM). The inner nuclear layer (INL) is perfused by the intermediate and the deep capillary plexa, which are supplied by vertical anastomoses from the second-order retinal arteries (A2) that give off precapillary arterioles orthogonally (Fig. 1B, asterisk). There is an oxygen tension (pO_2) gradient between the superficial inner retina and the MR layers and between the arteriolar and the venular ends. Between the territory supplied by two A2 arteries, the MR is at particular risk of hypo-oxygenation and hypoperfusion (watershed area W, Fig. 1B). Retinal vein occlusion will cause mainly perivenular ischemia [1] seen on *en face* OCT as a fern-like pattern of hyper-reflectivity. Distal precapillary ischemia [2] will cause more severe ischemic injury with extension into the superficial inner retina seen on *en face* OCT as focal areas of hyper-reflectivity (globular pattern). Transient or true major arteriolar occlusion [3] will appear on *en face* OCT as a band-like MR hyper-reflectivity mirroring the distribution of the occluded arteriole (arteriolar pattern). In addition to ischemia, reperfusion injury may also contribute to increased capillary dropout and PAMM [not depicted]. (Fig. 1B adapted from *McLeod*, 2019) [17]

tractional retinal detachment [4], vitreous hemorrhage [48], or epiretinal membrane removal [49]) and extraocular (pterygium removal [50]) ophthalmic procedures. Periocular anesthesia used in this case might be responsible for macular hypoperfusion in the setting of external ocular surgery [50]. PAMM after uneventful phacoemulsification surgery may occur due to spasm or transient occlusion of the central retinal artery (CRA) [46, 47]. Proposed contributing factors include vasculopathic risk factors or cerebrovascular disease, periocular anesthesia causing increased orbital pressure and reducing flow through the CRA [51] and increased intraocular pressure (IOP) in the immediate-to-early postoperative period. In a recent study, the incidence of PAMM after pars plana vitrectomy due to proliferative diabetic retinopathy was 3.8% [4]. Finally, eye compression injury causing global ocular ischemia has also been found to be causative of PAMM by compromising the blood flow in the CRA and posterior ciliary arteries [38].

PAMM in systemic diseases

PAMM has been identified in several systemic conditions, including neurologic/neuro-ophthalmic diseases such as idiopathic intracranial hypertension [52], meningitis [53], and leptomeningeal infiltration by tumors [54]. Carotid arterial disease has also been associated with PAMM, including internal carotid artery dissection and high-flow carotid-cavernous fistulas [54, 55]. Vascular surgeries such as endovascular coil embolization [56], aortic aneurysm repair [57], and cardiopulmonary bypass [58] have also been linked to PAMM. Iatrogenic embolic events following septoplasty with intraoperative use of triamcinolone acetonide have also been reported to lead to PAMM [59]. Typical PAMM signs have also been described in association with Susac syndrome [60], livedo reticularis [61], antiphospholipid syndrome [62, 63], and giant-cell arteritis [10, 27].

Vasculopathic risk factors may be related to PAMM, including SAH or hypotension, dyslipidemia [64], medications (including oral contraceptives), migraine disorder, anemia, and caffeine consumption [1, 2, 17]. There has been an increasing number of reports of drug-induced PAMM. The vasodilation caused by PDE-5 inhibitors may exacerbate the nocturnal dip in systemic blood pressure and cause transient retinal hypo/nonperfusion at the DCP [9, 65]. Sumatriptan intake for the treatment of migraine has been associated with PAMM [66]. The pathogenesis in this setting is likely multifactorial, including migraine-induced vasospasm and vasoconstriction at the DCP induced by the drug through selective 5-hydroxytryptamine 1d receptor agonist activity [66]. Finally, the performance-enhancing drug synephrine, a sympathomimetic with potent vasoconstrictive action through its partial agonist activity at alpha-adrenergic receptors, may cause PAMM by inducing retinal vasospasm [67].

PAMM in otherwise healthy patients

PAMM has been reported as a cause of visual symptoms in otherwise healthy patients [1, 2]. In some patients, a history of recent viral flu-like disease is elicited [1]. A case series of patients with PAMM has suggested that post-H1N1 vaccination status may be a risk factor [38]. An anecdotal case of PAMM in an otherwise healthy pregnant woman with no risk factors has been reported [23]; the authors proposed that occult branch retinal artery occlusion or vasospasm may have been possible contributing factors in this case [23].

Clinical features

PAMM can be unilateral or bilateral [13]. Patients with PAMM typically present with sudden onset of one or multiple paracentral scotomata [2, 13]. Patients can complain of blurred central vision or difficulty in focusing [13]. The visual acuity (VA) may be either normal or slightly decreased [1, 2, 13]. The mean age at presentation is 49–53 years, with no gender predilection [9, 14, 22]. Two studies reported a mean duration of symptoms before presentation of 8 days (ranging from hours to 30 days) [14, 68]. In five large series of eyes with PAMM, the reported presenting VA ranged from hand motion to 20/20, with the mean VA being between 20/42 and 20/120 [9, 14, 18, 22, 68]. Of note, 50–72% of eyes had a VA ≥ 20/50 at presentation, but 25–45% of eyes had a VA at presentation ≤20/200 [14, 18, 22, 68].

Fundus examination may be grossly normal [2]. In some cases, deeper, smoother, and grayer lesions have been described [2].

Imaging in PAMM

Spectral-domain optical coherence tomography (SD-OCT) and optical coherence tomography angiography (OCT-A)

In the acute phase, PAMM appears as a hyper-reflective band spanning the INL [40]. Over time, PAMM lesion progresses to atrophy, with INL thinning on SD-OCT [11, 12, 29]. Figure 2 depicts a case of PAMM in a patient with inferior branch retinal artery occlusion. Table 1 summarizes the main reported findings in case series of PAMM using SD-OCT, swept-source OCT, and OCT-A.

In the CAMM variant, the process also progresses from the acute stage of a hyper-reflective band in the INL to the chronic phase with INL and plexiform layer thinning [24]. In FH, this atrophic stage has the appearance of a “pseudo-foveal pit.” In addition, subtle disruptions of the ellipsoid and interdigitation zone can be observed in the chronic phase; these changes in the outer retina reflect areas of reduced photoreceptor density demonstrated by adaptive optics imaging [24].

OCT-A technology has made it possible to visualize the extent of ischemia in the ICP and DCP in the acute and chronic phases of PAMM [1, 12, 71].

En face OCT

Three patterns of distribution of PAMM have been identified using a combination of *en face* OCT technology and OCT-A [22]. These patterns are associated with the mechanisms of retinal microvascular ischemia (Fig. 1) [3, 11, 28, 40]. First, the arteriolar pattern follows a sectorial distribution coherent with transient occlusion of a large retinal arteriole with rapid restoration of blood flow [3, 22, 28]. Second, the globular pattern is seen as one or multiple ovoid patches of hyper-reflectivity and may be due to precapillary arterial occlusion related to SCR or embolic events [3, 22]. Finally, a fern-like pattern of hyper-reflectivity may be observed in CRVO due to perivenular ischemia resulting from stasis and hypo-oxygenation of the total retinal circulation [3, 22, 28, 31]. Of note, fern-like PAMM demonstrates a spectrum of variation consistent with the gradient and severity of ischemia in CRVO [68].

Swept-source optical coherence tomography

Swept-source OCT (SS-OCT) offers some advantages over SD-OCT, including faster scanning speed, high-detection efficiency, increased sensitivity with image depth, reduced fringe washout, longer imaging range, higher penetration into the optic nerve head and choroid, and lower sensitivity to ocular opacities [70]. En face swept-source OCT-A (SS-OCTA) has a superior depth and field of view compared with en face SD-OCTA [70].

On SS-OCT, the band-like hyper-reflective lesion in the middle retinal layers may show undulations at its margins, which reflects reperfusion and resolution of the infarction [13, 69]. The angiographic extent of the disease measured by SS-OCTA may be correlated with the confluent areas of INL atrophy seen on *en face* slabs as well as with visual field defects [13, 32, 39, 53, 70]. In addition, the repeatability of SS-OCTA allows observation of the dynamic course of the disease [59].

Fluorescein angiography (FA)

FA cannot clearly distinguish the ICP from the DCP [12]. Nevertheless, it may show findings related to the underlying etiology [12, 30, 31, 36, 60]. Published reports of PAMM have shown various findings on FA, including no vascular changes [3, 14, 28, 35, 42, 46], delayed filling in the areas corresponding to the PAMM lesions [7, 13, 40, 67], increased intracapillary spacing, and small areas of hypofluorescence [37]. In a case of chorioretinopathy secondary to juvenile dermatomyositis, bilateral retinal perivenous leakage was documented [44]. In the CAMM variant, FA showed delayed branch retinal artery filling, background diabetic retinopathy,

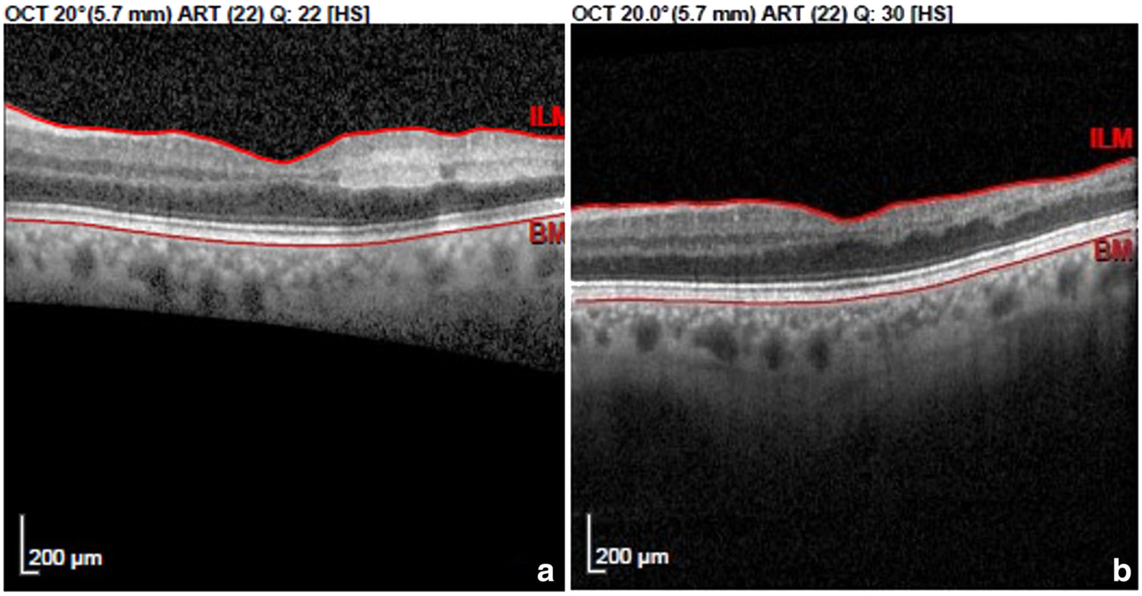

Fig. 2 Spectral-domain optical coherence tomography (SD-OCT, Spectralis, Heidelberg) of paracentral acute middle maculopathy (PAMM) in a 72-year-old hypertensive patient presenting with inferior branch retinal artery occlusion. In the acute phase (Fig. 2A), a hyper-reflective band spanning the inner nuclear layer (INL) was observed in the inferior retina. At the 12-week follow-up, progression to INL atrophy and thinning of the inferior retina was observed (Fig. 2B)

leakage without neovascularization, and indistinguishable boundaries of the FAZ [24].

Indocyanine green angiography (ICGA)

We only identified two reports of ICGA findings in patients with PAMM. The first published angiographic evidence of retinal ischemic changes responsible for PAMM showed occlusive changes in a third-order retinal artery that appeared as a late intraluminal vascular hyperfluorescence on ICGA [17]. However, in another report of PAMM, ICGA changes were related to the baseline etiology [42].

Infrared reflectance imaging

Infrared reflectance imaging has been shown to be a useful imaging tool for detecting PAMM and AMN lesions. PAMM lesions have been reported as dark areas in the macular area corresponding to SD-OCT lesions [34, 72], but the findings may be unremarkable [28]. In the CAMM variant, a foveal hyporeflective area was observed, which colocalized with the lesions seen on autofluorescence imaging [24].

Blue light fundus autofluorescence (FAF)

Blue light fundus autofluorescence findings in PAMM have been reported in a case of Purtscher's retinopathy as hypoautofluorescent lesions corresponding to fundus lesions located in the parafovea corresponding to fading Purtscher flecken [34], but another report regarding PAMM found no changes using this modality [3]. In the CAMM variant, FAF showed a globular-shaped hypoautofluorescent lesion [24].

Electrophysiology

Full-field electroretinography (ERG) in PAMM may show changes related to the underlying etiology (e.g., CRVO or CRAO), but the PAMM lesion should not be detected by this method. However, multifocal ERG may show decreased amplitudes in the affected areas [3, 35, 62].

Microperimetry

In the acute phase, microperimetry (MP) may disclose relative scotomas mapping to the areas of band-like hyper-reflectivity in the middle retinal layers. Persistent paracentral scotomata that colocalize with the areas of INL atrophy in chronic PAMM can be documented in serial MP testing [22, 30, 62, 65, 73].

While areas of DCP changes in PAMM are related to areas of decreased retinal sensitivity in MP, no study has analyzed whether the degree of retinal sensitivity loss in PAMM is correlated with the extent of ischemia and DCP capillary dropout. However, areas of more extensive DCP capillary dropout on OCT-A are correlated with lower retinal sensitivity on MP in patients with diabetic macular ischemia and in SCR [74, 75].

Table 1 Diseases and conditions associated with paracentral acute middle maculopathy in the literature as determined by spectral-domain or swept-source optical coherence tomography or optical coherence tomography angiography imaging

First author (year of publication)	Technology	Eyes (n)	Baseline diagnosis	OCT findings	OCT-A findings	Presenting symptoms/signs	Visual outcomes	Additional findings	Comments
Rahimy et al. (2014) [14]	SD-OCT Spectralis, Heidelberg	25 (5.2% of studied eyes)	Acute nonischemic CRVO	Hyper-reflective plaque-like lesions involving the INL	N/R	Central visual deficits Mean presenting VA = 20/120 (range: 20/20-HM)	Mean VA improved to 20/40 (mean follow-up time = 7.2 months); 82% of eyes with VA ≥ 20/40 had persistent scotomas	No correlation to FA lesion	–
Nemiroff et al. (2015) [21]	SD-OCT (Optovue, Inc., Fremont, California, USA)	8 eyes (4 acute and 4 chronic cases of PAMM)	CRAO, BRAO, SCR, trauma.	Hyper-reflective lesions in the middle retinal layers (acute cases) Thinning of the INL (chronic cases)	Normal OCT-A in acute cases; Pruning in the DCP in chronic cases (mean reduction in DCP vessel density = 19.4%)	Paracentral scotoma Mean VA = 0.46 ± 0.64 logMAR (3 eyes with normal VA, but 1 eye CF)	Improvement in scotoma was reported in only 1 patient	–	–
Dansingani et al. (2015) [27]	SD-OCT Spectralis, Heidelberg	4 eyes (2 patients)	Acute CILRAO; Background of GCA [1] Branch retinal vein occlusion occlusion [1]	Hyper-reflective lesion in IPL, OPL and INL, followed by thinning of the GCL, IPL and INL 1 month later	Reduced flow in the DCP and in the SCP	Unchanged VA in both cases, with paracentral scotomas	N/R	Reduced flow in DCP in the acute stage	PMH: GCA in 1 patient
Casalino et al. (2016) [12]	SS-OCT-A; AngioVue OCT-angiography system, Optovue, Inc.	3 eyes (3 patients)	CRVO	Typical features of PAMM	DCP showing capillary dropout, perfusion deficits and pruning	2 eyes with normal VA; 1 eye with decreased VA 20/250	VA deteriorated in 1 eye (from 20/250 to 20/800). Persistent scotomas	Fundus had venous engorgement and a retinal hemorrhage at the posterior pole	Relevant PMH: arterial hypertension; dyslipidemia, transient ischemic attack (1 patient); myocardial infarction (1 patient)
Falavarjani et al. (2017) [68]	SD-OCT (Spectralis, Heidelberg Engineering, Heidelberg, Germany).	11 eyes (11 patients)	5 CRVO, 2 hemicentral retinal vein occlusion, 4 none	Hyper-reflective band-like in the level of INL Fern-like pattern of perivenular PAMM was identified in most patients on en face OCT	N/R	Mean presenting VA = 20/43 (range, 20/20 to CF)	Mean final VA improved to 20/31	FA pattern of significantly delayed and slowed arterial flow in 9/11 eyes	PMH: included hypertension, coronary artery disease, coagulopathy, chronic obstructive pulmonary disease, human immunodeficiency virus infection, hyperthyroidism and glaucoma
Creese et al. (2017) [47]	SD-OCT unspecified	10 eyes (10 Patients)	None	Hyper-reflectance of the middle macular layers	N/R	VA decreased to 3/36 to HM after surgery	VA outcome reported in 1 eye (no improvement in VA at 4-week follow-up)	–	8809 cataract operations were performed; the frequency of PAMM after cataract surgery was 0.068%
			None				N/R	–	En face fern-like pattern

Table 1 (continued)

First author (year of publication)	Technology	Eyes (n)	Baseline diagnosis	OCT findings	OCT-A findings	Presenting symptoms/signs	Visual outcomes	Additional findings	Comments
Al-Mamoori et al. (2018) [69]	SS- OCT DRI OCT Triton	2 eyes (2 patients)		Hyper-reflectivity of the IPL, INL and OPL.	Perifoveal capillary dropout in SCP and DCP	Decreased VA (range: 6/60 to CF); Scotomas			
Jung et al. (2018) [70]	SS-OCT unspecified	2 eyes (2 patients)	Sickle cell maculopathy	Multiple confluent areas of macular INL thinning	Decreased SCP and DCP perfusion	VA range from 20/20–20/25; Scotomas	N/R	–	SS-OCTA clearly demonstrated the angiographic extent of the disease correlating with the Humphrey visual field scotomas and confluent areas of INL thinning
Kulikov et al. (2018) [20]	SD-OCT Copernicus REVO- Version 7.2; Opto- pol, Zawiercie, Poland	6 eyes with PAMM; 2 eyes with acute macular neuroretinopathy (AMN)	None. 1 patient with AMN and PAMM following flu-like symptoms.	PAMM:		hyper-reflective bands within the INL (acute); plaque-like thinning of the INL (chronic).	PAMM: decreased vessel density in both SCP and DCP plexa	VA was normal in all PAMM cases; 2 eyes had paracentral scotoma	N/R
1 eye (PAMM)-FA reveled a few round hypofluorescent areas, which corresponded to the intraretinal hemorrhages	–								
Pichi et al. (2018) [41]	SD-OCT Spectralis HRA OCT; Heidelberg Engineering and Cirrus HD-OCT 4000, V.5.0; Carl Zeiss Meditec)	53 eyes (53 patients)	Cilioretinal artery obstruction (CILRAO): 52.8% isolated CILRAO; 43.4% with CILRAO secondary to CRVO; two patients with CILRAO due to biopsy-proven GCA	Hyper-reflective ischemia of both the middle and inner retina	N/R	Isolated CILRAO: mean VA = 20/50; Presence of paracentral (42.8%) or central (57.1%) scotoma CILRAO secondary to CRVO: mean VA = 20/40; GCA-CILRAO: VA 20/175	Isolated CILRAO mean final = VA 20/25. Patients with CILRAO secondary to CRVO normalized VA	–	–
Bakhoum et al. (2018) [18]	SD-OCT (Spectralis; Heidelberg Engineering, or Optovue, Fremont,	16 eyes (16 patients)	CRAO, BRAO, CILRAO, CRVO	Hyper-reflective band involving the INL	N/R	Mean presenting VA = 20/54 (range = 20/25 to HM)	Mean final VA improved to 20/42	Cotton wool spots were noted in the majority of cases	A perivenular fern-like PAMM pattern was associated with better visual outcomes (average final VA was 20/25)

Table 1 (continued)

First author (year of publication)	Technology	Eyes (n)	Baseline diagnosis	OCT findings	OCT-A findings	Presenting symptoms/signs	Visual outcomes	Additional findings	Comments
	California, USA, or CIRRUS HD-OCT, Carl Zeiss Meditec, Germany).								
Nakamura et al. (2018) [56]	Cirrus HD-OCT, Carl Zeiss Meditec AG	2 eyes (2 patients)	Endovascular coil embolization	Hyper-reflective bands in the INL	Variable changes in SCP and DCP capillary abnormalities; reduced DCP vascular density at 2-month follow-up	Normal VA, but paracentral scotoma	Persistent paracentral scotoma	FA showed delayed filling in the 2 parafoveal lesions. 2 weeks; however, the DCP vascular density was reduced after 2 months	Relevant PMH: systemic arterial hypertension, hyperlipidemia, Tension headache
Burnasheva et al. (2019) [9]	SD-OCT RTVue-XR Avanti (Optovue, Fremont, CA)	102 eyes (51 patients)	Systemic arterial hypertension	INL thinning with OPL disruption Chronic PAMM lesions were found 88.9% hypertensive patients and 16.7% healthy individuals	Vessel density in SCP of hypertensive patients and healthy individuals was 48.9 ± 2.4% and 47.8 ± 2.0%, respectively.	N/R	N/R	N/R	.
Moura-Coelho et al. (2019) [54]	SD-OCT Spectralis, Heidelberg	3 eyes (3 patients)	Presumed leptomeningeal inflammation; Internal carotid artery dissection (ICAD ICAD (1 eye with			CRAO-associated PAMM, 1 eye with PAMM)	Hyper-reflective band in the INL (onset) with progression to INL atrophy	Reduced inner retinal perfusion	Decreased VA (range: 20/100 to CF)
Improvement in VA at final follow-up (range 20/25–20/20-0)	Fundus examination was normal in one case, showed a grayish lesion in another case, and a cherry-red macular spot in another case. Fundus examination showed embolic inferior macular infarction	PMH: metastatic lung cancer under crizotinib (1 case); vasculopathic risk factors (2 cases) painful Horner syndrome							
Schofield et al. (2019) 63	SD-OCT Unspecified	6 eyes (3 patients)	Antiphospholipid syndrome	Disruption of outer plexiform layers	Focal deep capillary loss	2/6 eyes with reduced VA; all	1 patient reported improvement in scotoma	OCTA with focal deep capillary loss	PMH: arterial hypertension (1 case);

Table 1 (continued)

First author (year of publication)	Technology	Eyes (n)	Baseline diagnosis	OCT findings	OCT-A findings	Presenting symptoms/signs	Visual outcomes	Additional findings	Comments
						with recurrent scotomas			Exogenous estrogen use (all cases); Livedo reticularis (1 case)
Shah et al. (2019) [13]	SS-OCT, Atlantis, Topcon Corporation, Tokyo, Japan	2 eyes (2 patients)	Otherwise healthy patients	Hyper-reflectivity of inner retinal layers, which showed either an arteriolar or a fern-like pattern on en face OCT imaging	Capillary dropout in the DCP	Normal VA with paracentral scotoma	N/R	Whitish-gray intraretinal lesion at the fovea on fundus examination. Normal FA (1 eye) or blocked choroidal fluorescence in the area of retinal whitening (1 eye) FAF was either normal or with a hypoautofluorescent lesion corresponding to the fundus lesion	Multifocal electroretinography showing blunting of retinal responses correlated with retinal lesions
Michalak et al. (2020) [58]	SD-OCT unspecified	4 eyes (2 Patients)	Cardiopulmonary bypass	Hyper-reflectivity of the INL	N/R	VA range = 20/20–20/50; 1 eye with scotoma	1/4 eyes deteriorated VA to CF. 3/4 eyes recovered visual function	Scattered CWS were identified bilaterally	After cardiac surgery involving cardiopulmonary bypass
Maltsev et al. (2020) 33	SD-OCT RTVue-XR Avanti	123 subjects	45 BRVO, 21 CRVO, 57 controls	INL thinning (71.1% of BRVO, 71.4% CRVO, 19.3% healthy patients)	Flow signal attenuation or voids in chronic PAMM	N/R	N/R	N/R	INL thinning was found in both eyes in 25.0% with BRVO, in 40.0% of patients with CRVO and in 11.1% healthy patients. PAMM was highly prevalent in fellow eyes of patients with unilateral retinal vein occlusion

OCT optical coherence tomography, *OCT-A* optical coherence tomography angiography, *SD-OCT* spectral-domain optical coherence tomography, *SS-OCT* swept-source optical coherence tomography, *OD* right eye, *OS* left eye, *BRAO* branch retinal arterial occlusion, *VA* visual acuity, *OPL* outer plexiform layer, *INL* inner nuclear layer, *FA* fluorescein angiography, *CRVO* central retinal vein occlusion, *HM* hand motion perception, *SCR* sickle cell retinopathy, *CRAO* central retinal arterial occlusion, *GCA* giant-cell arteritis, *DCP* deep capillary plexus, *CF* counting fingers, *CWS* cotton wool spots, *CILRAO* cilioretinal arterial occlusion, *AMN* acute macular neuroretinopathy, *ICAD* internal carotid artery dissection, *GCL* ganglion cell layer, *PMH* past medical history, *VA* visual acuity, *PMMA* polymethyl methacrylate, *IPL* inner plexiform layer, *FAF* fundus autofluorescence imaging, *RAPD* relative afferent pupillary defect, *VFD* visual field defect, *N/R* not reported

Diagnostic workup and differential diagnosis

The presence of PAMM raises a wide differential diagnosis that requires a systemic workup to exclude potential infectious, inflammatory, vascular, toxic, and iatrogenic causes. A detailed history and physical examination should be performed in every patient presenting with PAMM. Past medical history should include inquiring about SAH, cardiovascular and cerebrovascular disease, diabetes *mellitus*, dyslipidemia, migraine, recent surgery, current medications, caffeine consumption, cigarette smoking and substance abuse, and associated signs and symptoms. A history of recent viral infection has been associated with PAMM in otherwise healthy patients [1, 22]. A comprehensive ophthalmic examination is essential to assess visual function and to identify signs of ocular disease, including uveitis, retinal vascular disease or vasculitis, or optic neuropathy.

Pertinent workup includes a complete blood count including a platelet count, prothrombin and activated partial thromboplastin times, serum glucose levels, and a lipid profile. Carotid ultrasound imaging and Doppler may be considered to detect stenosis [68]. In the appropriate setting, screening for sickle cell disease, hypercoagulability states (serum homocysteine, factor V Leiden), and autoimmune diseases, including multiple sclerosis [76, 77], systemic lupus erythematosus, antiphospholipid syndrome, Behçet's disease, or giant-cell arteritis, should be considered [62, 63]. HIV-related microvasculopathy and opportunistic CMV retinitis have been reported to be causes of ischemic maculopathy [78, 79], and PAMM has been found in patients with HIV infection [68]. Other infectious agents may be considered in the differential diagnosis of PAMM, including tuberculosis, syphilis, herpesvirus, hepatitis B and C viruses, toxoplasmosis, cat-scratch disease, and Lyme disease [65]. An anecdotal case of PAMM in a patient with Rickettsia infection was recently reported (Arede et al., personal communication).

An important entity that must be distinguished from PAMM is AMN. Although AMN shares common aspects with PAMM, including risk factors (hypercoagulability states, oral contraceptives, smoking, or association with prodromal viral syndrome) [80] and clinical manifestations (paracentral scotoma with or without change in visual acuity), they have distinct morphologic features. On fundus examination, AMN typically presents with flat red-brown wedge-shaped lesions with the apex pointing towards the fovea in contrast with the deep grayish lesions observed in PAMM [81]. Both entities may appear without a visible fundus lesion [82]. On OCT imaging, in AMN, the hyper-reflective band in the acute phase is deeper than that in PAMM and is located at the level of the outer plexiform/outer nuclear layer junction; it evolves into attenuation of the ellipsoid zone and external limiting band [83].

The diagnostic workup should be individualized and take into account the patient's past medical history, presence of cardiovascular risk factors, and ocular and systemic findings (Table 2). In some patients, no cause can be identified even after extensive workup [1, 2].

Prognosis

There is still much to be understood about the natural course of PAMM. A number of patients have INL atrophy and persistent paracentral scotomas years after the first episode [1, 17], and visual acuity outcomes may be affected. However, spontaneous improvement has been described [11]. In four large studies of patients with PAMM, the mean VA improved from 20/43–20/120 to 20/25–20/42 at the final follow-up (mean follow-up between visits of 2.4–9.5 months) [9, 14, 18, 68]. Visual outcomes are also related to the etiology of PAMM and severity of ischemic injury as well as concurrent ocular comorbidities [14, 18, 68].

In patients with PAMM secondary to CRVO, the extent of DCP flow voids and the time to reperfusion may be related to the patient's visual outcome [5]. In addition, the *en face* OCT pattern of PAMM is correlated with the final visual acuity [18], with eyes with perivenular fern-like PAMM having better visual acuity than eyes with globular PAMM. This

Table 2 Proposed diagnostic workup to exclude systemic causes of PAMM and alternative diagnoses. Of note, diagnostic testing should be individualized and take into account the patient's history and ocular and systemic findings

Systemic workup
Complete blood cell count
Hemoglobin, hematocrit, and CBC including platelet count
Complete metabolic and lipid panel (serum triglycerides, serum cholesterol and HDL-c and LDL-c fractions), serum glucose, HbA1c)
Infectious agents
Toxoplasmosis, *Bartonella*, *Borrellia*, *Treponema pallidum,* CMV, HSV-1 and HSV-2, VZV, HBV, HCV, HIV-1, and HIV-2
Autoimmune and inflammatory markers
C-reactive protein, ESR, C3, C4, ANA, ANCA, and antiphospholipid antibodies
Other evaluations
Blood pressure, carotid Doppler ultrasound, primary care physician review

CBC complete blood count, *HDL-c* high-density lipoprotein cholesterol, *LDL-c* low-density lipoprotein cholesterol, *HbA1c* glycated hemoglobin, *CMV* cytomegalovirus, *HSV-1* herpes simplex virus type 1, *HSV-2* herpes simplex virus type 2, *VZV* varicella zoster virus, *HBV* hepatitis B virus, *HCV* hepatitis C virus, *HIV-1* human immunodeficiency virus type 1, *HIV-2* human immunodeficiency virus type 2, *ESR* erythrocyte sedimentation rate, *ANA* antinuclear antibodies, *ANCA* antineutrophil cytoplasmic antibody correlation is strongly supportive of a continuum of ischemia extending laterally in the INL and DCP and anteriorly into the inner retina [18].

To date, there is no treatment shown to revert PAMM, and treatment should be targeted at the underlying cause of PAMM when present [5].

Conclusions

Paracentral acute middle maculopathy is an SD-OCT finding of high clinical significance that likely represents an underlying process of inner retinal ischemia. The development of SD-OCT and OCT-A technologies has allowed further understanding of this process. It has been shown to be associated with retinal vascular conditions; more importantly, however, this finding has provided an explanation for visual disturbances in otherwise healthy patients, including those with a normal fundus examination. This highlights the importance of ocular multimodal imaging in the diagnosis of retinal diseases and in patients with unexplained visual symptoms.

Literature research We searched the PubMed and Google Scholar databases to identify articles published up to 30 April 2020, using the following key terms: paracentral acute middle maculopathy, PAMM, and macular ischemia. In addition, we manually searched the reference lists of most primary articles.

Compliance with ethical standards

Conflict of interest All authors declare that they have no conflicts of interest related to this paper.

Ethical approval This article does not contain any studies with humans or animals performed by any of the authors.

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